

CELL DIVISION PROTEINS OF ARCHAEA

Julian GATZKI¹, Sonja-Verena ALBERS², Maria HOERNKE¹

¹ Martin-Luther-Universität, Physical Chemistry, Halle, Germany;

² Albert-Ludwigs-Universität, Biology, Freiburg.Br., Germany

(corresponding author: Maria.Hoernke@chemie.uni-halle.de)

Keywords: peripheral membrane proteins, membrane binding domain, archaeal membranes, monolayers, infrared spectroscopy

ABSTRACT

Archaea have developed an entirely different way of constructing their membranes compared to bacteria and eukarya, able to withstand harsh conditions (temperatures, salt, pH). Apart from general membrane behavior, we study the membrane binding of the cell division protein SepF and its binding partners [1]. Archaeal lipid monolayers are examined by surface-sensitive infrared spectroscopy to yield mechanistic information. Furthermore, vesicles are used for biophysical characterization including fluorescence spectroscopy for membrane packing and membrane permeabilization and for the local organization of lipids and bound proteins. The project is funded by the European Social Fund Plus (ESF+) together with the Federal State of Saxony-Anhalt (Grant No. ZS/2024/09/189789 to MH).

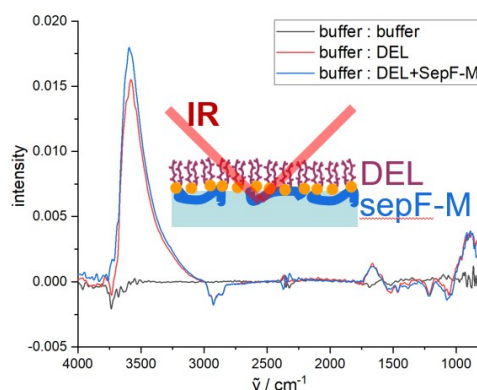


Fig. Infrared reflection-absorption spectra of the membrane-binding domain (*sepF-M*) of the cell division protein *SepF* interacting with a monolayer of archaeal diether lipids (*DEL*)

REFERENCES

- 1 Nußbaum, P.; et al. *Nat Commun*, **12** (1), 34692021, (2021)